

stion ber	Answer	Additional guidance	Mark
	(a series of) changes in the (types of) species (within a habitat) with time (1)		(1)

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	The correct answer is B (a place that meets the environmental conditions as organism needs to survive) <i>A is incorrect because it describes a community of organisms</i> <i>C is incorrect because it describes a population</i> <i>D is incorrect because an organism's niche is its role</i>	(1)

Question number	Answer	Mark
)	Early changes: • bare rock needs breaking down • by pioneer species / presence of alder and spruce • soil conditions improve when organisms die Increase in numbers of plants: • alder / spruce one of first plants to appear • because early soil is suitable / early stages of succession have to take place • because seeds brought in • numbers of alder / spruce rise • because the plants reproduce • alder reproduces faster than spruce • because there is little competition from other plants • hemlock last of the three plants to appear • because seeds brought in later • because early soil is not suitable for this plant • numbers of hemlock increase • because of reproduction Numbers of hemlock become stable: • because only this number can be supported by the environment • because still too many spruce trees present Decrease in number of plants: • numbers of alder / spruce fall • due to competition for {water / light / mineral ions / space} • due to disease • number of alder fall / drop to zero • because spruce out competes alder for {water / light / mineral ions / space} • because spruce is taller / grows faster • affected more by the appearance of hemlock • spruce numbers do not fall to zero • numbers of hemlock are not high enough to completely out compete the species	(6)

			Additional guidance
I 0	0	No awardable content	
I 1	1-2	An explanation may be attempted but with limited interpretation or analysis of the scientific information and with a focus on mainly just one piece of scientific information. The explanation will contain basic information, with some attempt made to link knowledge and understanding to the given context.	Simple descriptions / limited explanation of the data 1 mark = a description of some changes 2 marks = a simple explanation for one aspect of data OR an account of succession with no reference to the changes shown
I 2	3-4	An explanation will be given, with occasional evidence of analysis, interpretation and/or evaluation of both pieces of scientific information. The explanation shows some linkages and lines of scientific reasoning, with some structure.	Some explanation of the changes 3 marks = simple explanations for two aspects of data 4 marks = simple explanations for three aspects of data
I 3	5-6	An explanation is made that is supported throughout by sustained application of relevant evidence of analysis, interpretation and/or evaluation of both pieces of scientific information. The explanation shows a well-developed and sustained line of scientific reasoning, which is clear and logically structured.	Detailed explanation of the changes 5 marks = plus one additional detailed explanation 6 marks = plus two additional detailed explanations

